

**Spatio-Temporal Modeling of PM2.5 in
Berlin, Germany**

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Problem Setting

- High concentrations of PM2.5 is linked to respiratory issues, and governments take action to reduce the amount in the air.
- Being able to get a probabilistic look-ahead is useful for policy-makers and citizens to identify higher-risk areas.
- Some SOTA approaches are done at the INWT consulting firm where NO₂ was modeled for the state of Berlin using XGBoost.

Data Set

- I utilize sds011 sensor PM2.5 time series data from sensor.community
- The data is sufficient in many instances for my forecasting models though lots of preprocessing is done:
 - Daily averaging
 - Quantization/regularization
 - Omission of locations with more than 10 missing values
- Importantly, the dataset is dynamic and grows as time moves on
 - This leads to online deployment issues that must be solved

Methods

Workflow:

Sensor Community -> Ingest -> Aggregate -> Forecast + Interpolate -> Frontend

- **Ingest:** Pulls data from sensor.community and stores them in a PVC
- **Aggregate:** Uses Apache Spark to funnel the data into a parquet
- **Forecast:** Uses Chronos2 to make zero-shot temporal forecasts
- **Interpolate:** Uses Ordinary Kriging to interpolate between sensors
- **Frontend:** Displays the modeled PM2.5 map in a Streamlit window

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- Ingest and Frontend are deployed on the BHT data science cluster.
 - Aggregate, Forecast, and Interpolate use GPU, and are run on demand.

Results

- The main (viewable) result is the demonstrator that forecasts PM2.5
- Accessible here: <http://pm25.project.ris.bht-berlin.de/>
- Additionally, the pipeline serves as an example of online-deployed ML
- Note, the maximum PM2.5 per 24 hour period is: 35 $\mu\text{g}/\text{m}^3$ (EPA)

Discussion

- The demonstrator provides a way to examine the different areas of Berlin on different days of forecast.
- It extends the sensor.community work by:
 - Predicting a day in the future
 - Looking at the spaces between sensors
- It looks at a different air-quality measure than the SOTA INWT project
 - So it doesn't overlap completely, rather serves as a supplement
- It is a novel contribution to the analysis of the Berlin environment

Questions?